

Systematic Evidence Gap Analysis: Absence of Spacetime Scale Constraints in Allowed Corpus

Input Question

What is the smallest scale of space-time?

Domain

Astronomy

Validation Status

needs_data

Problem Statement

The exact value of the smallest scale of space-time has not been empirically determined or theoretically agreed upon within the provided evidence corpus. Current physical laws break down at scales such as the Planck length, making meaningful measurements problematic. The provided evidence set consists of documents from computer science, chemical physics, mathematics, and graph theory, none of which contain direct information regarding quantum gravity or spacetime granularity.

Rationale

Determining the fundamental scale of space-time is crucial for unifying quantum mechanics and general relativity. However, the allowed evidence IDs (EV-Q058-141657d189f4bf479ce9f855, EV-Q058-ba62c330ee7d6f08be4185f6, EV-Q058-60878f944b636e4fcfda99bb, EV-Q058-2f6a24cf545bf8aa4c95fc69) are domain-mismatched. This research plan aims to formally verify this absence of evidence through a Systematic Evidence Gap Analysis, distinguishing semantic overlaps (e.g., 'string' in CS vs. Physics) from physical constraints.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Within the closed set of allowed evidence (EV-Q058-141657d189f4bf479ce9f855, EV-Q058-ba62c330ee7d6f08be4185f6, EV-Q058-60878f944b636e4fcfda99bb, EV-Q058-2f6a24cf545bf8aa4c95fc69), there exists no empirical or theoretical basis to determine the smallest scale of space-time, constituting a verified negative result rather than a physical finding.
- **Mechanism**: The determination of a fundamental spacetime scale requires evidence from high-energy physics or quantum gravity. The provided evidence corpus consists exclusively of unrelated domains: string attractors in computer science (EV-Q058-141657d189f4bf479ce9f855), chemical physics of solids (EV-Q058-ba62c330ee7d6f08be4185f6), geometric lattice surfaces (EV-Q058-60878f944b636e4fcfda99bb), and

graph Laplacians (EV-Q058-2f6a24cf545bf8aa4c95fc69). Therefore, any assertion regarding spacetime scales derived solely from this set is epistemically invalid. This hypothesis treats the 'undeterminability' as a conditional property of the dataset, not a universal physical law.

- ****Falsifiable Prediction****: If a systematic review protocol applied to the allowed evidence IDs identifies at least one document containing valid physical constraints on spacetime granularity (e.g., Planck scale derivations, Lorentz violation bounds), this hypothesis of 'dataset-specific undeterminability' is immediately falsified.
- ****Required Observations****: Systematic verification that EV-Q058-141657d189f4bf479ce9f855 refers to Burrows-Wheeler Transform compressibility, not String Theory ; Confirmation that EV-Q058-ba62c330ee7d6f08be4185f6 addresses condensed matter/chemical physics without QG implications ; Verification that EV-Q058-60878f944b636e4fcfda99bb and EV-Q058-2f6a24cf545bf8aa4c95fc69 are purely mathematical constructs unrelated to physical spacetime metrics
- ****Risk of Being Wrong****: Low risk within the constrained system boundaries; the primary risk is semantic ambiguity where terms like 'string' or 'lattice' are misinterpreted as physical concepts. This is mitigated by strict domain classification protocols.

Hypothesis 2

- ****Hypothesis****: Semantic overlap between information-theoretic 'strings' (EV-Q058-141657d189f4bf479ce9f855) and physical 'string theory' creates a false positive signal for spacetime scale determination, which must be formally excluded via domain-specific ontology mapping.
- ****Mechanism****: Automated retrieval systems may conflate 'smallest string attractor' (a measure of repetitiveness in text compression) with 'fundamental string length' in quantum gravity. This hypothesis posits that the apparent relevance of EV-Q058-141657d189f4bf479ce9f855 is an artifact of polysemy. Validating the absence of spacetime information requires explicitly distinguishing these ontologies to prevent category errors in evidence synthesis.
- ****Falsifiable Prediction****: If EV-Q058-141657d189f4bf479ce9f855 is shown to contain discussions linking Burrows-Wheeler Transform metrics to physical spacetime discreteness or quantum gravity phenomenology, the 'polysemy artifact' hypothesis is falsified.
- ****Required Observations****: Ontological distinction between 'string attractor' (CS) and 'fundamental string' (Physics) in the text of EV-Q058-141657d189f4bf479ce9f855 ; Absence of cross-domain citations linking the paper's metrics to Planck-scale physics
- ****Risk of Being Wrong****: Moderate risk; interdisciplinary work occasionally bridges these fields, though no such bridge is indicated in the provided excerpt. Failure to detect a subtle link would invalidate the exclusion.

Technical Details

This research plan constitutes a Systematic Evidence Gap Analysis (SEGA) conditioned strictly on the provided allowed evidence set. The objective is to formally verify the absence of empirical or theoretical constraints on the smallest scale of spacetime within the corpus defined by IDs: EV-Q058-141657d189f4bf479ce9f855, EV-Q058-ba62c330ee7d6f08be4185f6, EV-Q058-60878f944b636e4fcfda99bb, and EV-Q058-2f6a24cf545bf8aa4c95fc69. The methodology adapts PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines for automated evidence screening. Instead of physical experimentation, the 'experiment' is a rigorous domain ontology mapping and semantic exclusion protocol. Each evidence card is evaluated against a controlled vocabulary of Quantum Gravity (QG) and High-Energy Physics

(HEP) terms (e.g., 'Planck length', 'Lorentz violation', 'spacetime granularity'). The hypothesis posits that all four documents belong to disjoint domains (Computer Science/String Attractors, Condensed Matter/Chemical Physics, Mathematical Geometry/Lattice Surfaces, and Linear Algebra/Graph Theory) and thus contain zero information regarding spacetime metrics. Validation requires demonstrating that no semantic bridge exists between these specific mathematical constructs and physical spacetime discreteness in the provided texts.

Datasets

Source

```
```json
```

```
[
```

```
{
```

```
"id": "EV-Q058-141657d189f4bf479ce9f855",
```

```
"description": "ArXiv paper on string attractors and Burrows-Wheeler Transform compressibility (Computer Science).",
```

```
"domain_label": "Information Theory"
```

```
},
```

```
{
```

```
"id": "EV-Q058-ba62c330ee7d6f08be4185f6",
```

```
"description": "ArXiv paper on chemical physics of solids and innovative measurement technology (Condensed Matter).",
```

```
"domain_label": "Chemical Physics"
```

```
},
```

```
{
```

```
"id": "EV-Q058-60878f944b636e4fcfda99bb",
```

```
"description": "ArXiv paper on lattice surfaces and smallest triangles in geometric contexts (Mathematics).",
```

```
"domain_label": "Geometric Topology"
```

```
},
```

```
{
```

```
"id": "EV-Q058-2f6a24cf545bf8aa4c95fc69",
```

```
"description": "ArXiv paper on grounded Laplacian eigenvectors and graph theory bounds (Linear Algebra).",
```

```
"domain_label": "Graph Theory"
```

```
}
```

]

'''

## Target

Binary classification matrix: {Evidence\_ID} -> {Contains Spacetime/QG Constraints?}. Expected output: All False.

## Paper Abstract

Background: The smallest scale of space-time remains undetermined, with current laws breaking down at the Planck scale. Objective: To verify whether the provided evidence corpus contains any empirical or theoretical constraints on this scale. Methods: We conducted a Systematic Evidence Gap Analysis on four allowed evidence IDs, applying domain ontology mapping and semantic exclusion protocols to distinguish physical spacetime concepts from mathematical analogies (e.g., string attractors, lattice surfaces). Results: Pending execution of validation experiments. Conclusion: Preliminary analysis suggests the corpus is domain-mismatched, containing no relevant quantum gravity literature. This study formalizes the verification of this negative result.

## Methods

1. Ontology Mapping: Define strict exclusion criteria based on the absence of QG-specific terminology. 2. Semantic Disambiguation Protocol: Explicitly distinguish polysemous terms (e.g., 'string' in CS vs. Physics). 3. Systematic Exclusion Review: Apply modified PRISMA flow to document screening reasons. 4. Negative Result Verification: Confirm the intersection of allowed evidence and required spacetime observations is empty.

## Experiments

### Baselines

```json

[

"Domain Expert Consensus Baseline: Manual verification by a physicist that the listed arXiv categories (cs.DS, cond-mat, math.GT, math.SP) do not typically contain primary results on quantum gravity spacetime scales.",

"Keyword Absence Baseline: Verification that standard QG keywords (Planck, Lorentz, Graviton, Discretization) have zero frequency in the provided text excerpts."

]

'''

Metrics

```json

[

"Exclusion Precision: Percentage of evidence cards correctly identified as non-relevant to spacetime physics (Target: 100%).",

"Semantic Ambiguity Score: Measure of potential confusion between mathematical analogies (e.g., lattice surfaces) and physical models (Target: 0, indicating clear distinction).",

"Gap Completeness Index: Binary metric indicating whether all allowed evidence IDs have been screened and documented as irrelevant (Target: 1)."

]

'''

## Ablation

Test the robustness of the exclusion by checking if removing any single evidence ID changes the conclusion (it should not, as all are irrelevant). Test sensitivity to title-only vs. abstract-only vs. full-text analysis to ensure no hidden relevance exists in deeper sections.

## Validation Protocol

Cross-reference the domain labels assigned to each EvidenceCard with their actual arXiv subject classes (if available in metadata) or explicit content quotes. Ensure that the 'verification of absence' is documented as a systematic review output, not a statistical inference.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q058-141657d189f4bf479ce9f855\*\*** · arxiv · arXiv:2506.05638
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2506.05638.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:3|section:page-3|paragraph:1;  
content\_sha256=42589760e48b79187e231cf9c4b03ec6f6afa097169e8ffcf35d838543408a9a
- **\*\*EV-Q058-ba62c330ee7d6f08be4185f6\*\*** · arxiv · arXiv:2301.01368
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2301.01368.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=a2322864116d9e753ba1c21ab6553ed586d0fd0478b18762a734ca6222b6f50d

- **\*\*EV-Q058-60878f944b636e4fcfda99bb\*\*** · arxiv · arXiv:1512.00908
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1512.00908.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=ae78199be64fab9b593de6e6de08a2aab5e55db08c9fd042a26360a52a92d281
- **\*\*EV-Q058-2f6a24cf545bf8aa4c95fc69\*\*** · arxiv · arXiv:1406.2271
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1406.2271.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=5c519f5dc5b3b21078c94bf881d87c606326db2f9248dde0d84d0ddfa29825d7

## Reviewer Comments

- The revised hypothesis correctly reframes the investigation as a 'Systematic Evidence Gap Analysis' conditioned strictly on the allowed evidence set, resolving the previous category error.
- Baselines have been appropriately updated from NLP metrics to systematic review criteria (Domain Expert Consensus, Keyword Absence), aligning methodology with the negative-result nature of the study.
- The mechanism explicitly limits the scope of 'undeterminability' to the provided corpus, avoiding overgeneralization to universal physics.
- Results field correctly states 'pending' and specifies verification of evidence absence rather than physical measurement.
- All supporting evidence IDs are valid and correctly mapped to their respective non-physics domains in the dataset description.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Verify that only the four specified EvidenceIDs are included in the screening pool.
- Document the exact list of excluded QG keywords used for the absence check.
- Provide the rationale for domain classification for each ID (e.g., quoting the 'Burrows-Wheeler' context for EV-Q058-141657d189f4bf479ce9f855).
- Ensure no external physics literature is cited as evidence for the gap; the gap is defined solely by the provided corpus.

- Confirm that the output is structured as a Negative Result Report/Systematic Gap Analysis.